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BYOND

VC Liquidity Scorecard

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Executive Summary

This project develops a liquidity scorecard to help Byond LLC identify venture capital firms that are most likely to have available capital for investments. Founders often spend valuable time reaching out to investors who may not be actively investing, which slows the fundraising process. The goal of this analysis is to provide a targeted, data-driven way to identify investors with a higher likelihood of deploying capital.

To address this problem, we compiled a dataset of venture capital firms focused in the medtech, healthtech, and digital health spaces. Data was collected from sources such as Crunchbase, PrivCo, firm websites, and news articles with validation and source tracking to ensure accuracy. Key indicators included investment frequency, recent fund activity, follow-on behavior, fund age, and capital deployment.

We applied Gradient Boosting and Random Forest models to predict investor liquidity and choose indicators, respectively. The Gradient Boosting performed the best of any model we tested on the data we have, achieved strong accuracy, and the highest ROC-AUC. This makes it the most practical model for building a transparent scorecard.

The results show that the strongest indicators of liquidity are whether a firm has announced a new fund in the last twelve months, and its average number of deals per year. These variables consistently signal active capital deployment.

The liquidity scorecard can be used to rank investors and guide outreach efforts, allowing founders to focus on the most relevant targets. This approach improves efficiency and increases the likelihood of successful fundraising outcomes.

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Background

Byond LLC is a growth advisory firm that works with startups in the areas of medtech, healthtech, digital health, and SaaS. The firm helps founders raise money for their businesses. At the seed and early Series A stages of a startup, founders need capital to grow both their products and their businesses. In the case of venture capital, a startup raises money from a venture capital firm or an angel group. These groups invest money in the startup in exchange for equity or ownership in the company. However, not all venture capital firms are able to invest money at all times. Because of this, the fundraising process can often be slow and uncertain for founders. It can be difficult to know which investors are most likely to be actively investing at a given time. Having a clearer way to evaluate investor activity could make the fundraising process more efficient and help founders focus more time and resources on building their companies.

Problem

The main problem addressed by this project is identifying which investors currently have capital available to invest. Venture capital firms often appear active based on their reputation or past investments, but that does not necessarily mean they currently have available liquidity. Because of this issue, founders often reach out to a large number of investors during the fundraising process. Some of these investors may not be investing at the time, which can slow the process down and make fundraising more difficult. This also takes time away from other important work, such as developing products or growing the company. To help address this issue, the project aims to develop a liquidity scorecard that evaluates investors based on observable signals including recent fundraising activity, investment patterns, hiring trends, and portfolio outcomes. By analyzing these factors, the scorecard will help estimate which investors are more likely to have capital available for new investments and therefore be more relevant targets for founders seeking funding.

Data

The data used for this project will come from exclusively publicly available data sources related to venture capital activity. Because venture capital activity is not centralized in a single dataset, multiple sources must be used to build a more complete picture of the industry. These sources provide information about venture firms, their investments, and their portfolio companies.

As an initial step, the project team compiled a list of venture capital firms that are active in digital health and related sectors. Examples include firms such as Rock Health Capital, Flare Capital Partners, Define Ventures, General Catalyst, Andreessen Horowitz Bio + Health, Khosla Ventures, Lux Capital, and Bessemer Venture Partners. All these firms have the common connection of investing in the medtech, healthtech, and digital health space. After discussion with Byond, the exclusion of Andreessen Horowitz Bio + and other firms which do not have a portfolio focus on the medtech, healthtech, and digital health space was conducted. However, after further discussion with Byond the scope of firm investments changed once again and firms like Andreessen Horowitz (firms which have broad portfolios but still invest in the specified sectors) were allowed again. These firms will serve as a set of observations for the analysis of this project. Data will be collected on characteristics such as investment frequency, portfolio growth, fund age, round amount raised, firm follow-on, and other characteristics for analyzing investor behavior and liquidity signals.

We employed a system prompt (Appendix E.1) to interact with Gemini 3 for web scraping. Preprocessing is simple with the prompt receiving a text file containing a list of firms to search for. The prompt is then executed via Anaconda Prompt and interacts with the Gemini 3 API to update a JSON file with the

relevant information. The resulting file serves as the structured dataset used for subsequent analysis and model development.

Guardrails are put in place to ensure that the LLM does not hallucinate firm names or information provided. For example, in an early iteration, the model incorrectly indicated that 7wireVentures had invested in “Family.cards.” Upon further investigation using Crunchbase and PrivCo, we found that the model had mistakenly conflated OHA Osnabruck 7 Ventures with 7wireVentures.

We implemented additional guardrails (See Appendix E.2 for example or the ZIP file for whole conversation) to ensure data quality and reliability. We instructed the model to only pull information from reputable sources such as the firm’s website, the SEC, Crunchbase, PrivCo, PitchBook, and credible news articles. The model also returns source links in which the information was found, allowing us to manually verify the data. Upon discussion with Byond, every single collected indicator for every single firm has a matching link (or links) which point to where the information was found. Additionally, we limited the output to return only Seed and Series A investments from 2022 through 2026. The rationale for selecting this time frame is as follows:

Including data from too far back introduces a major external shock, which is COVID-19, which would skew the results and reduce their reliability. A limitation of this approach is that fund-level data is not consistently available for every firm. Therefore, the model is designed to be directionally accurate rather than perfectly precise.

Information collected includes, but is not limited to, last fund open date, average round size, follow-on share percentage, total exits from 2022 to present, last fund size, years since last fund closed, and related measures. Team members took ten individual investments at random and verified the information provided against known, reliable sources such as Crunchbase and PrivCo. All the information collected was deemed accurate. The columns collected and their methodology can be found in Appendix A, Table 1.

A note on the data for this project. As of 04/07/26, we were given a full list of 200 firms which Byond would like us to pull and do analysis on. We attempted to pull the data, but when doing so we ran into two issues. One being that the servers seemed to be constantly overloaded (it turns out Gemini 3 seems to be deprecated) and two being the JSON file not truncating properly. After fixing issue two, and upgrading to Gemini 3.1 for the model, we pulled the data. This data, however, was not trustworthy. Thus, we made the decision to combine two old datasets for a total of 193 firms.

This was done by taking the oldest dataset and aggregating the data upward, as the oldest dataset was not at the same level of detail as the more recent dataset, then combining the two and removing duplicates. The method of aggregation can be found in Appendix B, Table 2.

Methodology

The overall project workflow is shown in Appendix D, Figure 3.

Our methodology focuses on evaluating liquidity-related signals across a set of firms and their investment activity. We began by collecting data via a system prompt scraper, as was discussed earlier, and the data captured included investments, funding, portfolios, company characteristics, and other organizational attributes relevant to the liquidity assessment. We collected the data at the firm level.

We based this approach on the premise that liquidity cannot be directly observed but can be indirectly observed or inferred by analyzing investment behavior and the behavior of the firm more broadly. Firms that invest more frequently and recently are more likely to have available capital, while firms with limited or outdated activity may be between fundraising cycles or have already deployed most of their capital.

To determine which indicators are most important, we used preliminary feature importance results to identify which variables are most strongly associated with investment activity. These results guided our selection of indicators to include in the model. As our research progresses, we compare how each indicator

aligns with observed investment activity.

In order to collect the data in a proper manner, specific prompts and guardrails were used. For instance, one of the initial prompts which specified firm names looked like so:

“All information pertaining to the investments these VCs are making ought to come from reputable sources. The VCs’ own website, news articles, Crunchbase, PrivCo, etc. You ought not make interpretations with a firm’s name; for instance, ‘7wireVentures’ is not ‘OHA Osnabruck Healthcare 7 Ventures.’ Additionally, all companies these firms are making investments in ought to be in the healthcare space. Specifically medtech, healthtech, and digital health.”

Ultimately, this is what it looks like in the Python file which Gemini reads:

STRICT RULES — follow these exactly:

1. Use the firm name as given: “{firm}.” Minor formatting variations are acceptable — spacing differences, capitalization, or the addition or omission of a legal suffix (LLC, Inc., Partners, Capital, etc.).
2. What is *not* acceptable is interpreting, translating, or substantially renaming the firm — do not expand an acronym, translate a word, or replace the name with a different organization entirely.
3. If you cannot find the firm under its name or any close formatting variant, return the structure with all numeric fields as `null` and counts as 0.

You can see the full prompt and reply in the Appendix E.2.

There are some limitations when it comes to this method of collecting data. First, there is some inherent uncertainty. It is entirely possible that somewhere in the data collected there is incorrect information. Second, only publicly available data can be collected in this specific manner. While you can use an API for specific sites to access private information in a similar manner, with the time allotted for this project and the stated goal of Byond being to collect as much data as possible via publicly available sources, we did not add private data to our scraper output.

When we received the data, we first verified that it captured the correct information. Because the data was originally collected at the fund and investment level, we could easily identify inconsistencies such as investments that did not align with Seed or Series A rounds, or those outside the medtech, healthtech, or digital health sectors. After refining the query, we were able to collect only the data which was specified and at the firm level.

Next, we validated the accuracy of the data. Team members reviewed ten randomly selected investments and cross-checked them against trusted and reliable sources such as PrivCo and Crunchbase. Though the data were clean in that they were all in the same format, some data were missing due to hiccups when collecting the data — this includes errors from Gemini operating at peak times. Thus, these results were removed and we had a total of 193 rows. The data was in a formatted JSON file which we used directly in Python for analysis.

We ran into an issue with the data. The data are heavily skewed in favor of liquid firms, based on the definition of liquidity which Byond outlined. Unfortunately, with the number of firms in the dataset being so small, we have seemingly no recourse with this.

We conducted a relatively simple exploratory analysis to ensure we had a balanced and even distribution of firms across all fund stages and to avoid skewed results. Based on the VC Funds Lifecycle document, most firms in our sample had funds which were considered young.

There were a few analytical steps to figure out which indicators are most useful for predicting liquidity. We created a liquidity label based on the definition of having invested in the last 12 months, and where the fund was in the fund lifecycle. This liquidity score was a weighted output defined as follows:

$$LS = 0.65 AI + 0.35 LW \quad (1)$$

The active investor (AI) variable is a dummy variable weighted at 65%. The lifecycle weight (LW) is determined by inferring the average liquidity across a fund's lifecycle, using the lifecycle document provided by Byond. For example, funds younger than 0.9 years have a weight of 0.8, funds between 0.9 and 5 years have a weight of 1.0, and funds that are between 5.1 and 8 years have a weight of 0.5.

To test which indicators are most important, we employed a classification model which used k -fold cross-validation over 5 splits of the data. Each variable was shuffled one time to assess how the results changed and how changes in that feature impacted model performance. Based on this analysis, the model focuses on a small set of indicators that capture recent investment activity, capital deployment, and consistency over time. These indicators include average round size, average deals per year, total exist 2022 to present, follow on share percentage, firm founded year, and last fund size in millions.

Model Details

While, given the nature of the dataset and the goal of this project, we hoped that logistic regression made the most sense for the investor liquidity scorecard, it turns out that Gradient Boosting Machine was the best model for this data. While we also explored models such as Random Forest and Logistic Regression, we found Gradient Boosting to be especially valuable because it produces results the most accurate. We prioritized accuracy because our client, Byond, needs to understand (1) which firms are likely to have available capital (liquidity prediction) and (2) which observable signals are driving that prediction. See Appendix F.1 for the H2O grid search. Model selection and other code can be found in the ZIP file.

The liquidity score is calculated as follows:

$$Y_i = 1[(\text{investments in last 12 months}_i > 0) \wedge (\text{current fund age in years}_i < 5)] \quad (2)$$

We used the H2O package in Python to run the Gradient Boosting. Much like choosing the indicators, liquidity is defined as a combination of having made an active investment in the last 12 months and fund age. Liquidity itself is not directly observable, so we rely on these measurable signals to approximate whether a firm has capital available to invest. Firms that have invested recently and are still within an active portion of their fund lifecycle are more likely to have deployable capital, whereas firms that have not invested recently or are later in their lifecycle are less likely to be actively investing. Fund age, however, is simply a binary of being below 5 years of age for the simplicity of the model and interpretability of the results. This threshold aligns with industry norms, where venture capital funds tend to deploy the majority of their capital within the first few years after raising a fund. While this simplification reduces some nuance from the lifecycle approach used earlier, it improves interpretability and ensures the results are easier to communicate in a practical setting.

The predicted probability that firm i is liquid given the characteristics of $\mathbf{x}_i$ is

$$\hat{p}_i = \frac{1}{1 + e^{-F(\mathbf{x}_i)}} \quad (3)$$

$$F(\mathbf{x}_i) = \beta_0 + \sum_{m=1}^M \nu h_m(\mathbf{x}_i) \quad (4)$$

$$\hat{Y}_i = \begin{cases} 1 & \text{if } \hat{p}_i \geq c \\ 0 & \text{if } \hat{p}_i < c \end{cases} \quad (5)$$

Here, $\mathbf{x}_i$ denotes the set of predictor variables, or indicators, used to evaluate firm i . These predictors include measures such as average deals per year, whether a new fund was announced in the last 12 months, total follow-on investments, follow-on share percent, firm founded year, average round size, and last fund size. In equation (3), $\hat{p}_i$ is the predicted probability that firm i is liquid, bounded between 0 and 1. This probability is obtained by applying the logistic transformation to the model score $F(\mathbf{x}_i)$.

Equation (4) defines that model score. The term β_0 is the initial baseline prediction, and the summation adds the contribution of many individual decision trees, $h_m(\mathbf{x}_i)$, each scaled by the learning rate ν . Rather than estimating one coefficient per variable, gradient boosting builds the prediction iteratively: each new tree is added to improve upon the errors made by the earlier trees. This allows the model to capture nonlinear relationships and interactions among the indicators that may not be well represented by a simple linear specification.

Equation (5) converts the predicted probability into a binary classification using a cutoff value c . If the predicted probability for a firm is greater than or equal to the cutoff, the firm is classified as liquid; otherwise, it is classified as not liquid. In this way, the model produces both a probability score and a final classification.

In laymans terms, gradient boosting makes a prediction by learning from its mistakes, which resembles a tree. The first prediction is looked at by the second prediction to learn why the prediction was too high or too low, and the third one looks at the second one in the same manner. Say you have a student whose final exam score you're trying to guess. You make a guess, then you notice it's too low when you consider students who studied a lot so you adjust accordingly, then you notice it's too high when you consider students who skipped class, so you adjust accordingly again.

An important implication of using gradient boosting is that the effect of any one predictor is not summarized by a single coefficient. Instead, a variable may influence the prediction differently depending on the values of other variables and the branches of the trees in which it appears. For that reason, model interpretation relies more naturally on tools such as SHAP-based explanations rather than coefficient signs and magnitudes.

Because the dataset is relatively small at 193 firms, model complexity must still be controlled carefully. In this setting, gradient boosting is made more stable through the use of tuning parameters such as the learning rate, tree depth, and cross-validation. These choices help reduce overfitting while still allowing the model to capture meaningful nonlinear patterns in the relationship between firm characteristics and liquidity. A full comparison of model performance is reported in Appendix C, Table 3.

The results of our analysis show meaningful differences in how well each model predicts investor liquidity using the selected indicators. We evaluated three models, Logistic Regression (H2O GLM), Gradient Boosting Machine (GBM), and Random Forest, using accuracy, precision, recall, F1 score, and ROC-AUC. Considering multiple performance metrics is important because no single measure fully captures a model's classification ability.

Overall, the GBM model performed best across most metrics. It achieved the highest accuracy (0.7755), the highest precision (0.7857), the strongest ROC-AUC (0.8721), and an F1 score of 0.8000. These results

suggest that GBM provided the best overall balance between correctly identifying liquid firms and limiting false positives, while also showing the strongest ability to distinguish between liquid and non-liquid firms across classification thresholds.

The Random Forest model also performed well, with an accuracy of 0.7551, recall of 0.8889, and an F1 score of 0.8000. Its recall was higher than GBM's, indicating that it was better at capturing liquid firms when the priority is to avoid missing potential opportunities. However, its lower precision (0.7273) and ROC-AUC (0.8552) suggest that it was somewhat less effective than GBM at making consistently well-separated classifications.

The Logistic Regression model had the lowest overall accuracy (0.7143), precision (0.6757), F1 score (0.7812), and ROC-AUC (0.8485), but it achieved the highest recall (0.9259). This indicates that Logistic Regression was especially strong at identifying liquid firms, though it did so with more false positives than the other models. Its results suggest that, while it remains useful and interpretable, it does not perform as strongly overall as the tree-based models in this application.

In summary, all three models show predictive value, which indicates that the selected indicators are informative for identifying investor liquidity. However, GBM provides the strongest overall performance, while Logistic Regression offers the greatest interpretability and the highest recall. For a transparent and explainable liquidity scorecard, Logistic Regression remains attractive, but for pure predictive performance, GBM appears to be the strongest model among those tested. The confusion matrix for the Gradient Boosting model is provided in Appendix C, Figure 1.

To further evaluate model performance, we analyzed the confusion matrix for the Gradient Boosting model. The model correctly identified 22 liquid firms and 17 non-liquid firms. It incorrectly classified 6 non-liquid firms as liquid and missed 5 liquid firms. This indicates that the model is more likely to classify firms as liquid, which aligns with its relatively high recall. From a practical perspective, this means the model is more likely to identify potential investors, even at the cost of including some firms that may not currently be liquid. The relative importance plot is provided in Appendix C, Figure 2.

We also examined the relative magnitude of the model effects to understand which variables were most influential in predicting liquidity. As shown in Figure 3, average deals per year is by far the strongest predictor, followed by whether a new fund was announced in the last 12 months and total follow-on investments. Last fund size and average round size also contribute, but to a much smaller extent. By contrast, firm founded year and follow-on share percentage appear to have very limited influence on the model's predictions.

These results suggest that recent and sustained investment activity is more informative for identifying liquidity than older firm characteristics or portfolio share measures. In particular, firms that invest more consistently and have recently raised new capital appear to be more distinguishable from less liquid firms. Overall, the figure reinforces the idea that liquidity is most closely associated with current market activity and fundraising momentum rather than static background characteristics.

Conclusion

This project set out to help Byond LLC solve a practical fundraising problem: identifying which venture capital firms are most likely to have capital available for new investments. Using publicly available data on 193 firms across the medtech, healthtech, and digital health space, we developed a liquidity scorecard that estimates investor liquidity from observable signals related to recent investment behavior, fund activity, and firm characteristics.

The results show that investor liquidity can be approximated with meaningful predictive value. Among the models tested, Gradient Boosting Machine performed best overall, achieving the strongest balance across accuracy, precision, F1 score, and ROC-AUC. The analysis also showed that average deals per year was the most influential predictor of liquidity, followed by whether a firm had announced a new fund in the last 12

months and, to a lesser extent, follow-on investment activity. Together, these findings suggest that liquidity is most closely associated with recent and sustained market activity rather than with static background characteristics alone.

For Byond and its clients, the scorecard offers clear practical value. Rather than relying primarily on reputation, brand recognition, or outdated investor signals, founders can use a data-driven ranking to prioritize firms that appear more likely to be actively deploying capital. This can reduce time spent on low-probability outreach, improve the efficiency of the fundraising process, and support more targeted advisory recommendations.

At the same time, the results should be interpreted with appropriate caution. The dataset is relatively small, limited to public data, and skewed toward firms classified as liquid. In addition, liquidity is not directly observable, so the model depends on a practical proxy based on recent investment activity and fund age. Future work should expand and refine the dataset, refine the liquidity definition, incorporate additional signals such as hiring trends or strategy changes, and, where possible, and integrate private data sources to improve model robustness. Even with these limitations, this project demonstrates that investor liquidity can be inferred systematically and translated into a practical scorecard that helps Byond deliver more focused, efficient, and data-driven fundraising support.

Appendix

A Indicator Definitions

Table 1: Definitions of liquidity indicators

Indicator	Definition
investments_last_12_months	Deals made in the most recent 12 months (from Mar. 2026)
avg_deals_per_year	Total investments divided by active years (2022–2026)
new_fund_last_12_months	Boolean—did they announce or close a new fund in the last 12 months
visible_down_rounds	Count of portfolio companies with known down rounds
avg_round_size_usd_millions	Mean round size across Seed + Series A (2022–2026)
follow_on_share_pct	Follow-ons as a percent of total investments
years_since_last_fund_close	Time since the last fund held its final close
current_fund_age_years	Age of the current or last fund from its open date
currently_raising_from_lps	Boolean—public evidence they are in market raising
pct_capital_deployed	Estimated percent of current fund deployed
total_exits_2022_present	Total exits from 2022 through today

B Field Mapping

Table 2: Mapping from old-format fields to new-format fields

New format field	Source
avg_deals_per_year	(investments_2022 + investments_2023 + investments_2024 + investments_2025 + investments_2026) / 5
investments_last_12_months	investments_2025 + investments_2026
avg_round_size_usd_millions	avg_investment_size_usd_millions
follow_on_share_pct	(follow_on / total_investments) * 100
years_since_last_fund_close	Calculated from last_fund_close_date → 2026-04-11
current_fund_age_years	Calculated from last_fund_open_date → 2026-04-11
new_fund_last_12_months	new_fund_in_market
currently_raising_from_lps	new_fund_in_market
visible_down_rounds	null (not in old format)
total_exits_2022_present	null (old had lifetime exits, not 2022-present)
pct_capital_deployed	null (not reliably derivable)
sources	Each field key maps to the old file's sources_used array

C Results Tables and Figures

Model	Accuracy	Precision	Recall	F1	ROC AUC
GBM	0.7755	0.7857	0.8148	0.8000	0.8721
Random Forest	0.7551	0.7273	0.8889	0.8000	0.8552
Logistic Regression (H2O GLM)	0.7143	0.6757	0.9259	0.7812	0.8485

Table 3: Model performance comparison

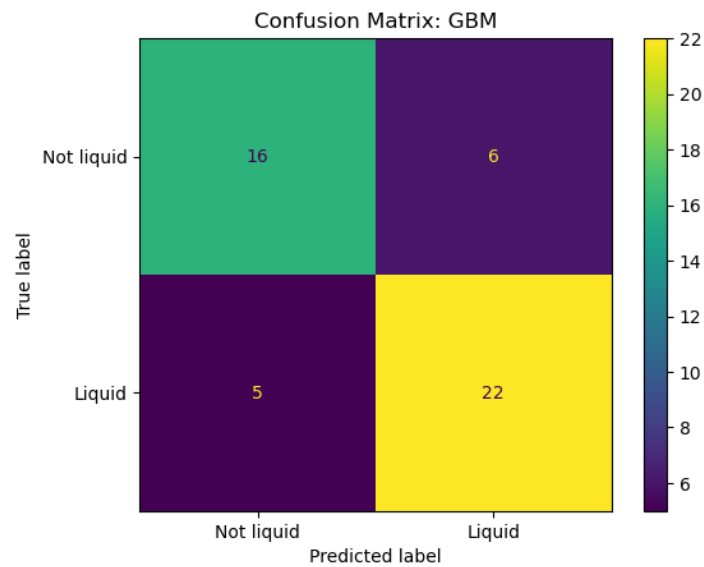


Figure 1: Confusion matrix for the GBM (H2O) model.

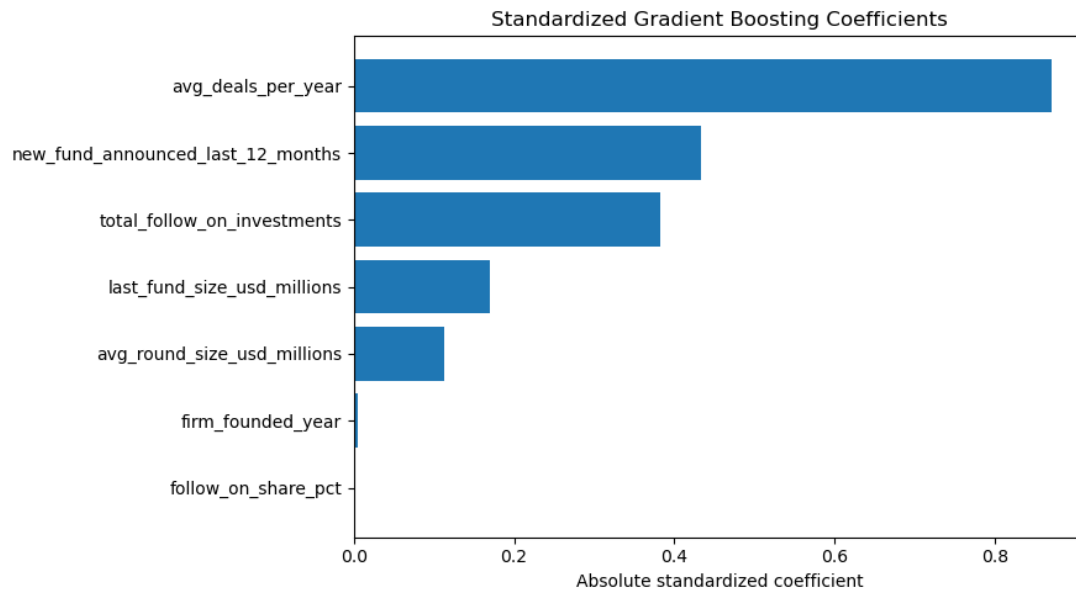


Figure 2: Absolute standardized gradient boosting coefficients.

D Additional Figures

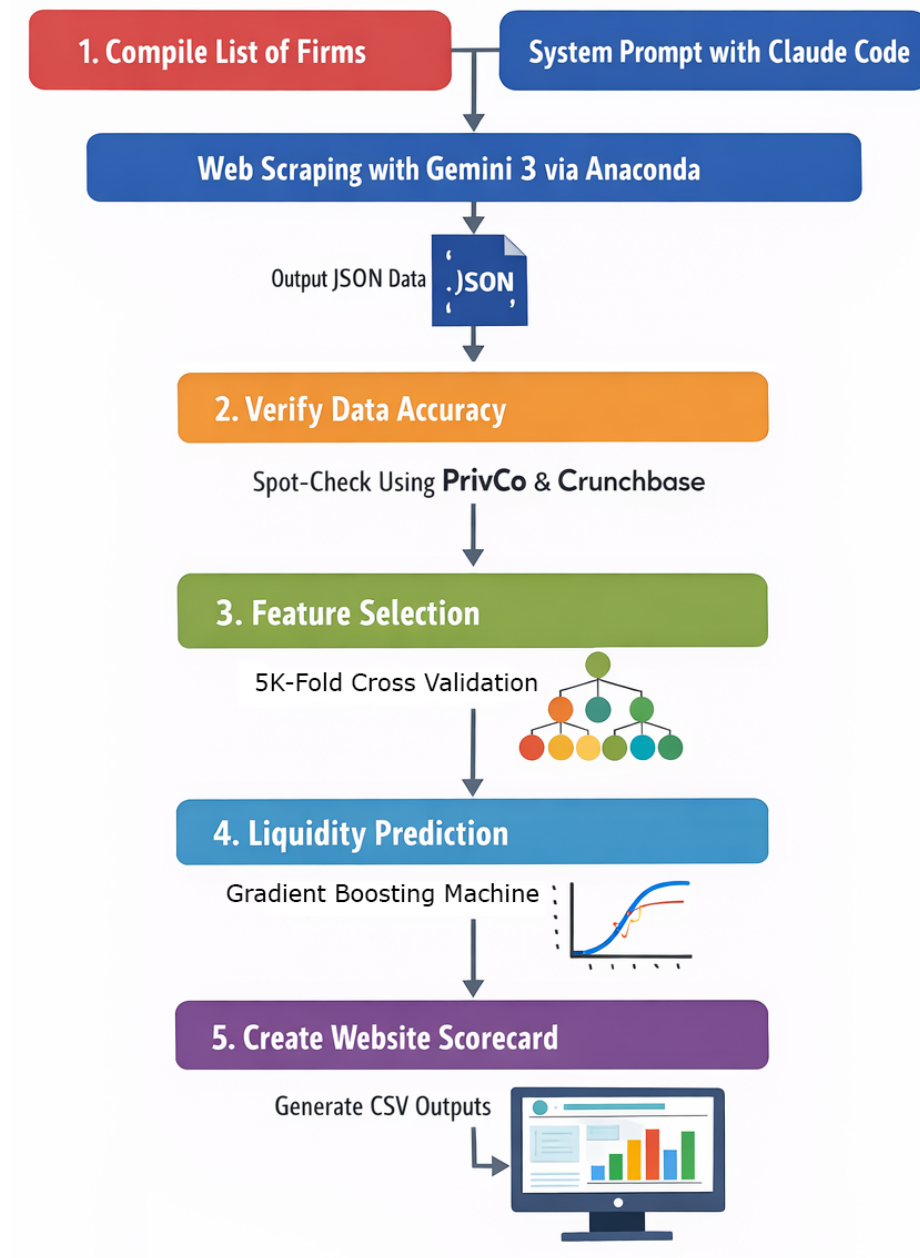


Figure 3: Project workflow.

E LLM Prompts

E.1 Gemini Prompt for Venture Capital Firm Search

The following prompt and code structure were used to build a system prompt that interacts with Gemini 3 to search the web. The prompt was designed to generate queries dynamically from a notepad document

containing a list of venture capital firms. Each generated query included, one by one, a firm name together with a keyword and year, such as “firm + Series A + 2020.” The intended output was a structured JSON object containing meaningful attributes about investments made by that firm.

I want to build a system prompt that interacts with Gemini 3 to search the web. I want to create the prompts dynamically based on a notepad document with a list of venture capital firms. The system prompt that it’s generated dynamically would include, one by one, a firm plus series A plus 2020, so firm plus keyword plus year. I want a structured JSON output with meaningful attributes about any investments that have been done to that firm.

To run this code, install the following dependency:

```
pip install google-genai

import os
from google import genai
from google.genai import types

def generate():
    client = genai.Client(
        api_key=os.environ.get("GEMINI_API_KEY"),
    )

    model = "gemini-3-flash-preview"
    contents = [
        types.Content(
            role="user",
            parts=[
                types.Part.from_text(text="\"INSERT_INPUT_HERE\""),
            ],
        ),
    ]
    tools = [
        types.Tool(googleSearch=types.GoogleSearch(
        )),
    ]
    generate_content_config = types.GenerateContentConfig(
        thinking_config=types.ThinkingConfig(
            thinking_level="HIGH",
        ),
        tools=tools,
    )

    for chunk in client.models.generate_content_stream(
        model=model,
        contents=contents,
        config=generate_content_config,
    ):
        print(chunk.text, end="")
```

```
if __name__ == "__main__":
    generate()
```

E.2 Gemini Prompt Guardrails for Source Quality, Firm Matching, and Healthcare Filtering

The following text shows an original prompt followed by a reply summarizing how that prompt was revised. The purpose of this exchange was to improve source quality requirements, preserve firm name integrity, and restrict results to healthcare-related investments only.

Original Prompt

All information pertaining to the investments these VC's are making ought to come from reputable sources. The VC's own website, news articles, Crunchbase, PrivCo, etc. You ought not make interpretations with a firm's name, for instance "7wireVentures" is not "Oha Osnabruck Healthcare 7 Ventures". Additionally, all companies these firms are making investments in ought to be in the healthcare space. Specifically medtech, healthtech, and, digital health.

Reply Summarizing Revisions to the Prompt

Done. Here's a summary of what was updated in the prompt:

Source enforcement

Must be verified by the firm's own site, Crunchbase, PrivCo, PitchBook, Bloomberg, Reuters, STAT News, MedCity News, etc.

Every investment must cite its `source_urls`

No inferring or hallucinating data — if it can't be verified, it's excluded

Firm name integrity

The model is explicitly told to use the firm name exactly as given — no translations, expansions, or interpretations

If the exact name yields no results, it returns an empty list rather than guessing

Healthcare-only filter

Only medtech, healthtech, or digital health companies are included

A `healthcare_subsector` field classifies each one

`top_sectors` renamed to `top_healthcare_subsectors` to reinforce the filter

Non-healthcare investments are fully excluded

F Code

F.1 H2O Grid Search

The following code was used to fit and compare Logistic Regression, Gradient Boosting Machine, and Random Forest models in H2O using grid search and cross-validated AUC.


```
glm_grid = H2OGridSearch(  
    model=H2OGeneralizedLinearEstimator(  
        family="binomial",  
        standardize=True,  
        lambda_search=True,  
        balance_classes=True,  
        nfolds=3,  
        fold_assignment="Stratified",  
        keep_cross_validation_predictions=True,  
        seed=42,  
    ),  
    hyper_params=glm_hyper_params,  
    grid_id=f"liquidity_glm_grid_{grid_suffix}",  
)  
  
glm_grid.train(x=x, y=response, training_frame=train_h2o)  
best_glm = glm_grid.get_grid(sort_by="auc", decreasing=True).models[0]  
  
gbm_grid = H2OGridSearch(  
    model=H2OGradientBoostingEstimator(  
        distribution="bernoulli",  
        balance_classes=True,  
        nfolds=3,  
        fold_assignment="Stratified",  
        keep_cross_validation_predictions=True,  
        seed=42,  
    ),  
    hyper_params=gbm_hyper_params,  
    search_criteria=gbm_search_criteria,  
    grid_id=f"liquidity_gbm_grid_{grid_suffix}",  
)  
  
gbm_grid.train(x=x, y=response, training_frame=train_h2o)  
best_gbm = gbm_grid.get_grid(sort_by="auc", decreasing=True).models[0]  
  
rf_grid = H2OGridSearch(  
    model=H2ORandomForestEstimator(  
        binomial_double_trees=False,  
        balance_classes=True,  
        nfolds=3,  
        fold_assignment="Stratified",  
        keep_cross_validation_predictions=True,  
        seed=42,  
    ),  
    hyper_params=rf_hyper_params,  
    search_criteria=rf_search_criteria,  
    grid_id=f"liquidity_rf_grid_{grid_suffix}",  
)
```

```
rf_grid.train(x=x, y=response, training_frame=train_h2o)
best_rf = rf_grid.get_grid(sort_by="auc", decreasing=True).models[0]

selected_models = {
    "Logistic Regression (H2O GLM)": best_glm,
    "GBM": best_gbm,
    "Random Forest": best_rf,
}

best_param_table = pd.DataFrame(
    {
        "model": list(selected_models.keys()),
        "model_id": [m.model_id for m in selected_models.values()],
        "cross_validated_auc": [
            m.model_performance(xval=True).auc() for m in selected_models.values()
        ],
        "main_parameters": [
            str(
                {
                    k: v
                    for k, v in m.actual_params.items()
                    if k in {
                        "alpha",
                        "lambda",
                        "ntrees",
                        "max_depth",
                        "learn_rate",
                        "sample_rate",
                        "col_sample_rate",
                        "mtries",
                    }
                }
            )
            for m in selected_models.values()
        ],
    }
).sort_values("cross_validated_auc", ascending=False)

best_param_table
```



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